

Cloud Computing Security Risks: Identification and Assessment

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Introduction

Cloud computing is arguably both an innovation in technology and an avenue for new business ventures. However, the revelations made by Snowden that the USA had been conducting mass surveillance and data collection through the US National Security Agency (NSA) and various other national intelligence agencies has created additional concerns about security when it comes to the cloud (Landau, 2013; Bauman, et al, 2014). As early as 2013 the German government adopted a particularly aggressive stance by seeking to mitigate the dangers of cloud technology by creating secure data centres in Germany specifically for email traffic. The use of SSL encryption was viewed as a way in which to restrict foreign jurisdictions from gaining access. It was on August 31, 2014, that a collection of almost 500 private pictures of various celebrities, mostly women, were hacked from the online storage offered by Apple's iCloud platform which is the source for automatically backing up photos from iOS devices, such as iPhones (Satti, 2015; Bai, Xing, Zhang, Wang, Liao, Li & Hu, 2017).

Cloud computing has been heralded as an innovation in information system architecture, with efficient usage of computer hardware resources (Zissis & Lekkas, 2010). However, with the exponential growth in the development and use of web based systems and computer technology brings with it an increased risk for security breaches from hacking (Monrose & Rubin, 1999; Choo, 2011; Teh, Teoh & Yue, 2013). The banking industry is a prime example of a high profile industry sector that has been the focus of the greatest number of attacks (Choo, 2011). In this regards the security of cloud based systems is also becoming a high risk prospect for cyber crime (Kaufman, 2009).

Subsequently, there has been a growth in the literature regarding risks and concepts for dealing with cloud security. For the most part it is important to bring these disparate concepts of the issues and the various approaches together in an effort to better

understand the recognition of the risks and the various methods for dealing with security. This paper is therefore concerned with drawing the information from the literature together in an effort to present a general overview.

Background

The National Institute of Standards and Technology (NIST) proposed a broad ranging definition of cloud computing and set out what they considered to be five essential features, three service models and four deployment models (Mell & Grance, 2011). The five essential features encompass; virtualized computing resource pool, broad network access, rapid elasticity, on-demand self-service, measured service. The three service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS); the four deployment models are private cloud, community cloud, public cloud and hybrid cloud.

With regards to the service models Infrastructure as a Service (IaaS) as the name implies involves a single tenant cloud layer where the Cloud computing vendor's dedicated resources are only shared with contracted clients at a pay-per-use fee. Software as a Service (SaaS) also operates on the pay-per-use costing model with software applications being leased out to contracted organisations by specialised SaaS vendors. Platform as a Service (PaaS) works on a similar basis to IaaS however, it provides an additional level of "rented" functionality. So there are different types of cloud computing services in much the same way as there are different types of clouds that exist in the sky¹.

A public cloud allows users' access to the cloud via interfaces using mainstream web browsers. Public clouds are less secure than the other cloud models because of the additional need to ensure that all applications and data accessed on the public cloud are not subjected to malicious attacks. A private cloud is established within an organisation and is therefore easier to align with the security, compliance, and regulatory requirements, providing greater control over deployment and use. ...

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